TIME & DISTANCE

1. speed $= \frac{\text{Distance}}{\text{Time}}$ $s = \frac{d}{t}$

$s \propto d$ [$\because$ t is constant]

$s \propto \frac{1}{t}$ [$\because$ d is constant]

$d \propto t$ [$\because$ s is constant].

$1 \text{ kmph} = \frac{1000 m}{60 \times 60} = \frac{5}{8}$ m/sec

x kmph $= x \times \frac{5}{18}$ m/sec

x m/s $= \frac{18}{5} \times x$ kmph

1. A train covers a certain distance in 210 min at a speed of 60 kmph. Find the time taken by the train to cover the same distance at a speed of 80 kmph.

$\boxed{\text{Ans: } 2\frac{5}{8} \text{ hrs}}$

210 min $= 3.5$ hrs $= \frac{7}{2}$ hrs.

$D = St = 60 \times \frac{7}{2} = 210$ km

$T = \frac{D}{S} = \frac{210}{80} = \frac{21}{8} = \frac{16}{8} \times \frac{5}{8} = 2\frac{5}{8}$ hrs

2. A part of journey is covered in 31.5 min at 80 kmph. And remaining part of journey in 75 kmph in 16 min. Find the total distance covered in the journey.

$\boxed{\text{Ans: } 62}$

$D = D_1 + D_2$

$= S_1 T_1 + S_2 T_2$

$= 80 \times \frac{31.5}{60} + 75 \times \frac{16}{60}$

$= 42 + 60$

$= 62$ km

3. The distance between 2 places A & B is 999km. An express train leaves from A at 6 am toward B ~~at~~ running at a speed of 55.5 kmph. The train stops on the way for 1hr 20 min. At what time it will reach B.

$\boxed{\text{Ans: 1hr : 20 min am}}$

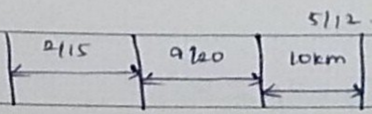
$T = (999 / 55.5) + 1hr\ 20\ min = 19\ hrs\ 20 min$

$= \left(\frac{999}{111} \times 2\right) + 1hr\ 20\ min = 19 hrs\ 20 min$ $\boxed{1:20\ am}$

4. A man covers 2/15 th part of total journey by train and 9/20 th part by bus and remaining 10km on foot. Find the total distance he covered in the journey.

$\boxed{\text{Ans: 24 km}}$

$\Rightarrow 1 - \left[\frac{2}{15} + \frac{9}{20}\right] \Rightarrow 1 - \frac{7}{12} \Rightarrow \frac{5}{12}$



$5/12 \longrightarrow 10 km$

$\underbrace{1}_{\text{Total distance}} \longrightarrow \frac{\cancel{10}^{2}}{\cancel{5}} \times 12 \Rightarrow 24\ km$

5. The speeds of 3 car are in the ratio 1:3:5. Find the ratio of time taken by the cars to cover the same distance.

$\boxed{\text{Ans: 15 : 5 : 3}}$

$\therefore S \propto \frac{1}{t}$ $\quad t \propto \frac{1}{S}$ $\quad t_1 : t_2 : t_3 = \frac{1}{S_1} : \frac{1}{S_2} : \frac{1}{S_3}$

$= \frac{1}{1} : \frac{1}{3} : \frac{1}{5}$

$= 15 : 5 : 3$

6) A certain distance is covered by cyclist in certain speed. If an jogger covers half the distance in 2 (double) the time, find the ratio of speed of jogger to that of cyclist.

$\boxed{\text{Ans: } 1:4}$

$\frac{1}{2}$ ~~of of jogger~~ =

	D	S	t
cyclist →	x km	S_1	y hrs
Jogger →	$\frac{x}{2}$ km	S_2	$2y$ hrs

$$S = \frac{D}{t}$$

$$\frac{S_1}{S_2} = \frac{D_1}{D_2} \times \frac{T_2}{T_1}$$

$$= \frac{x}{x/2} \times \frac{2y}{y}$$

$$= 2 \times 2$$

$$\frac{S_1}{S_2} = \frac{4}{1}$$

1

$[\therefore S_2 : S_1 = 1:4]$

7) The ratio of speeds of A & B is 3:5. If A takes 24 min more than B, to cover a certain distance then, so how much time B will take to cover the same distance.

$\boxed{\text{Ans: } 36 \text{ min}}$

$S_A : S_B = 3:5$ ~~The~~ [illegible]:

$T_A = (24+x)$ min

$T_B = x$ min

2

$$S \propto \frac{1}{T} \Rightarrow \frac{S_A}{S_B} = \frac{T_B}{T_A}$$

$$\frac{3}{5} = \frac{x}{24+x}$$

$$72 + 3x = 5x$$

$$2x = 72 \qquad \boxed{x = 36}$$

8) Two cars A & B travel from one city to another city 60 kmph & 108 kmph. If car B takes 2 hrs less time than car A for the journey. Find the distance b/w two cities.

$\boxed{\text{Ans: } 270 \text{ km}}$

$S_A = 60$ kmph $\qquad T_A = x$ hrs

$S_B = 108$ kmph $\qquad T_B = (x-2)$ hrs

$S \propto 1/t \qquad \frac{S_A}{S_B} = \frac{T_B}{T_A} \Rightarrow \frac{60}{108} = \frac{x-2}{x} \Rightarrow 5x = 9x - 18$

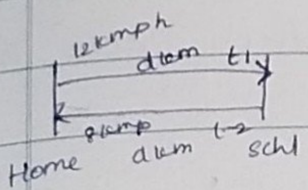
$x = 9/2$ hrs

$D = S_A T_A$

$= \overset{30}{\cancel{60}} \times \frac{9}{\cancel{2}} = 270$ km.

9) A boy went to his school at a speed of 12 kmph and returned to his home at a speed of 8 kmph. If he has taken 50 min for the whole journey. What is the total distance covered by him?

$\boxed{\text{Ans: } 8 \text{ km}}$



Total distance $= 2d$.

$t = t_1 + t_2$

$\frac{50}{60} = \frac{d}{12} + \frac{d}{8} \Rightarrow \frac{d}{4} = 1 \Rightarrow d = 4.$

$\therefore 2d = 8$ km

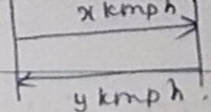
(or)

10) If a person goes from 1 place to another place with same time and with different speed,

$\text{Avg speed} = \frac{2xy}{x+y}$ kmph

$= \frac{2 \times 12 \times 8}{12+8} = 9.6$ kmph

$D = S \times T = 9.6 \times \frac{50}{60} = 8$ km.



10) If a train runs at a speed of 48 kmph. It reaches its destination late by 12 min. However if its speed is 64 kmph, it is late by 3 min only

what is the right time to cover the journey.

Ans:

Let actual time to reach is x min

$S_1 = 48$ kmph $\longrightarrow T_1 = (x+12)$ min $\qquad S \propto \frac{1}{t}$

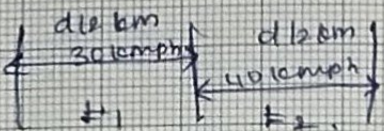
$S_2 = 64$ kmph $\longrightarrow T_2 = (x+8)$ min

$$\Rightarrow \frac{S_1}{S_2} = \frac{t_2}{t_1} \Rightarrow \frac{48}{64} = \frac{x+8}{x+12} \Rightarrow 3x+36 = 4x+12$$

$$\boxed{x = 24}$$

11) A car travels from A to B in 7 hrs. It covers half the distance in 30 kmph and remaining distance in 40 kmph. Find the distance b/w A & B

$\boxed{\text{Ans: } 240}$



$t = t_1 + t_2$

$7 = \frac{d/2}{30} + \frac{d/2}{40}$

$7 = \frac{d}{60} + \frac{d}{80}$

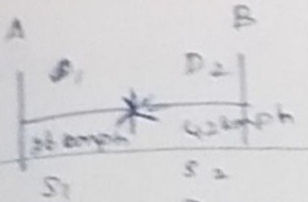
$= \frac{(80+60)d}{80 \times 60} = \frac{\cancel{140}^{7}d}{\cancel{80}_{4} \times 60} = \frac{7d}{240}$

$\cancel{7}d = 240 \times \cancel{7}$

$\boxed{d = 240}$

12) Two trains starts from A & B and proceeds towards B & A at 36 kmph, 42 kmph respectively. When they meet each other it is found that 1 train has covers 48 kms more than the other train. Find the distance b/w A & B?

$\boxed{\text{Ans: } 624 \text{ km}}$



(Time constant)

$S \propto D$

$D_{A-B} = x + x + 48$

$\frac{S_1}{S_2} = \frac{D_1}{D_2}$

$= 288 + 288 + 48$

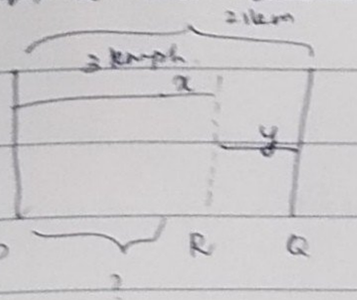
$= 624$ km

$\frac{36}{42} = \frac{x}{x+48} \Rightarrow \boxed{x = 288}$

13. 2 men A & B walk from P to Q a distance of 21 km at 3kmph & 4 kmph respectively. B reaches Q and returns immediately and meets A at R. Find the distance b/w P & R

$\boxed{\text{Ans: 18 km}}$

P, Q, R are distance



$D_A = D_{P-R} = 18$ km

$D = D_A + D_B$

$= 2(x+y)$

$= 2 \times 21 = 42$ km

$S_A : S_B = 3 : 4$

$D_A : D_B = 3 : 4$

$D_A = x$ km

$D_B = (x+y+y)$ km

$7p \to 42$ km

$1p \to 6$ m

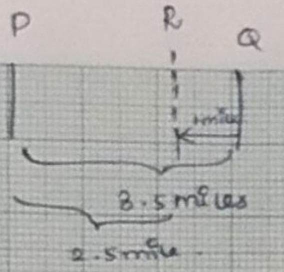
$D_A = 3p = 18$ km

14) B starts 4.5 mins after A from the same point for a place at a distance of 3.5 miles from the starting point. A on reaching the destination turns back and walks a mile where he meets B. If A walks 1 mile in 6 min. Then in how many mins B walks a mile?

$\boxed{\text{Ans: 9 min}}$

1

(wipro model question)



$D_A = 4.5$ miles $D_B = 2.5$ miles

$D_A \Rightarrow$ 1 mile $\rightarrow$ 6 min

4.5 mile $\rightarrow$ 4.5×6

27 min

$D_B \Rightarrow$ 2.5 miles $t_B = 27 - 4.5 = 22.5$ min

2.5 miles $\xrightarrow{B}$ 22.5 min

1 mile $\xrightarrow{B}$ $\frac{22.5}{2.5} = 9$ min

15) A train running at 7/11 th of its own speed reaches a place in 22 hrs. How much time could be saved if the train runs at its own speed?

$\boxed{\text{Ans: 8 hrs.}}$

$S \rightarrow T$

$\frac{7}{11} S \rightarrow 22$ hrs $S \propto \frac{1}{t}$

$\frac{S_1}{S_2} = \frac{T_2}{T_1}$ $\frac{S}{\frac{7}{11} S} = \frac{22}{T_1}$ $T = 14$ hrs

Time saved $= 22 - 14 = 8$ hrs.

16) while going at 4/5 th of his usual speed a man reaches to a place, 25 min late. what is the usual time taken by the man to reach that place.

$\boxed{\text{Ans: 100 min}}$

$S \propto \frac{1}{t}$

$\frac{S_1}{S_2} = \frac{T_2}{T_1}$

$\frac{S}{4/5\, S} = \frac{x + 25}{x}$

$\frac{5}{4} = \frac{x+25}{x}$

$5x = 4x + 100$

$\boxed{x = 100}$

17. A boy walking at a speed of 20 kmph reaches the school 15 min late. next time at the speed of 25kmph, he reached his school 10 min early. What is the usual time taken by the boy to reach his school.

$\boxed{\text{Ans: 110 min}}$

Let usual time to reach school = x min.

$S_1 = 20$ kmph $\qquad T_1 = (x+15)$ min.

$S_2 = 25$ kmph $\qquad T_2 = (x-10)$ min.

$$S \propto \frac{1}{t}$$

$$\frac{S_1}{S_2} = \frac{T_2}{T_1}$$

$$\frac{\cancel{20}^{4}}{\cancel{25}_{5}} = \frac{x-10}{x+15} \Rightarrow 4x+60 = 5x-50$$

$$\cancel{154} = \boxed{110 = x}$$

18. Walking at 60% of his usual speed, a man reaches his destination, 1 hr 40 min late. Find his usual time to reach his destination?

$\boxed{\text{Ans: } 2\frac{1}{2} \text{ hrs}}$

$S \longrightarrow T \qquad$ 1 hr 40 min $\Rightarrow 1 + \frac{40}{60} \quad 1 + \frac{2}{3} = \frac{5}{3}$ hrs.

$S_1 \quad S \longrightarrow T$ hrs

$S_2 \quad \frac{3}{5}S \longrightarrow \left(T + \frac{5}{3}\right)$ hrs

$$S \propto \frac{1}{T}$$

$$\frac{S_1}{S_2} = \frac{T_2}{T_1}$$

$$\frac{S}{\frac{3}{5}S} = \frac{T+\frac{5}{3}}{T}$$

$$5T = 3T + 5$$

$$T = 5/2$$

$$T = 2\frac{1}{2} \text{ hrs.}$$

19. A person covers a certain distance with certain speed. If he increased his speed by 15 kmph, then he will be 16 min early. By how much time he will be late if he reduces his speed by 12 kmph. If his initial speed is 60 kmph.

$\boxed{\text{Ans: 20 min}}$

60 kmph → T min

75 kmph → (T-16) min } T → $\frac{S_1}{S_2} = \frac{T_2}{T_1}$

$\frac{60}{75} = \frac{T-16}{T} \Rightarrow 60T = 75T - 75\times16$

$\frac{16}{15}\times 16 = \frac{3}{15}T$ (sic)

$\boxed{T = 80 \text{ min}}$

60 kmph → 80 min

48 kmph → T_2 (late)

$\Rightarrow \frac{60}{48} = \frac{T_2}{80}$

$\Rightarrow T_2 = 100$ min

⟹ He is late by $\underbrace{20 \text{ min}}_{[100-80]}$

20. Speed of B is 63.63% more than A. A & B walks for 36 hrs & 55 hrs respectively. If A moves 48 kms less than B. Find the total distance covered by A & B.

$\boxed{\text{Ans: 112 km}}$

$63.63\% \rightarrow \frac{7}{11}$

$S_A = 100\% = 1$

$S_B = 100\% + 63.63\%$

$= 1 + \frac{7}{11} = \frac{8}{11}$

$S_A : S_B = \left[1 : \frac{18}{11}\right] \times 11$

$= 11 : 18$

$t_A : t_B = 36 : 55$

$D_A : D_B = S_A T_A : S_B T_B$

$= \frac{11\times36}{18\times55} = \frac{2}{5} = \underbrace{2 : 5}_{3P}$

3P → 48

1P → 16

$D_A + D_B$

21. If a ma[n] missed ... walks ... station ... Find ... station.

$3p \rightarrow 48\ km$

$1p \rightarrow 16\ km$

$D_A + D_B = 7p = 7 \times 16 = 112\ km.$

21. If a man walks at a speed of 5 kmph. He missed the train by 7 min. However if he walks at a speed of 6 kmph he reaches the station, 5 min before the arrival of the train. Find the distance he covers to reach the station.

$\boxed{\text{Ans: 6 km}}$

Let T is usual time in min

~~kmph~~ ~~5 min~~

S_1 5 kmph (T+7) min

S_2 6 kmph (T-5) min

$\frac{S_1}{S_2} = \frac{T_2}{T_1} \Rightarrow \frac{5}{6} = \frac{T-5}{T+7}$ $= 5T + 35$

$= 6T - 30$

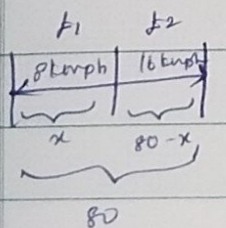
$\Rightarrow 65 = T$ $\quad T = 65$ min.

$D = S_1 T_1 = S_2 T_2$

$= 5 \times \frac{72}{60} \Rightarrow 6\ km$

22. A man travelled distance of 80 km in 7hrs. partly, on foot at 8 kmph and partly on cycle at 16 kmph. Find the distance he travelled on foot.

$\boxed{\text{Ans: 32 km}}$



$t = t_1 + t_2$

$7 = \frac{x}{8} + \frac{80-x}{16} = \frac{2x}{16} + \frac{80-x}{16}$

$\boxed{x = 32\ km}$

23 To cover a distance of 416 kms, train A takes $2\frac{2}{3}$ hrs more than train B. If the speed of train A is doubled, it would take $1\frac{1}{3}$ hrs less than B. Find the speed of train A.

$\boxed{\text{Ans: 52 kmph}}$

Let train B takes x hrs

$S_1 \rightarrow S_A \longrightarrow \left(x + \frac{8}{3}\right) \text{hrs} \rightarrow T_1$

$S_2 \rightarrow 2S_A \longrightarrow \left(x - \frac{4}{3}\right) \text{hrs} \rightarrow T_2$

$\frac{S_1}{S_2} = \frac{T_2}{T_1}$ $\qquad \frac{S_A}{2S_A} = \frac{x - 4/3}{x + 8/3}$

$x = \frac{8}{3} + \frac{8}{3}$

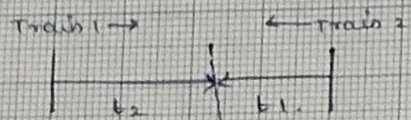
$x = \frac{16}{3}$ hrs

$T_A = \frac{16}{3} + \frac{8}{3} = \frac{24}{3} = 8$ hrs

$S_A = \frac{D}{T_A}$

$S_A = \frac{416}{8}$ $\boxed{= 52 \text{ kmph}}$

NOTES:

1. 

train 1 and train 2 are moving in opposite directions

$t_1, t_2 \rightarrow$ times taken by train-1 & train-2 to reach their destinations after meeting

$t \rightarrow$ time taken by two trains to meet each from starting time.

$t = \sqrt{t_1 t_2}$ $\qquad \frac{S_1}{S_2} = \sqrt{\frac{t_2}{t_1}}$

24. A & B started their journies from x, y to y, x respectively. After crossing each other, A & B reached their destination in $6\frac{1}{8}$ hrs and 8 hrs. If speed of B is 28 kmph. Find the speed of A.

$\boxed{\text{Ans: 32 kmph}}$

$$\frac{SA}{SB} = \sqrt{\frac{tB}{tA}}$$

$$\frac{SA}{28} = \sqrt{\frac{8}{49/8}} \Rightarrow \frac{SA}{28} = \sqrt{\frac{64}{49}} \Longrightarrow \frac{SA}{28} = \frac{8}{7} \Rightarrow SA = \frac{8 \times \cancel{28}^{4}}{\cancel{7}} = 32$$

SA = 32 kmph.

25. Two persons A & B travelling in opposite direction from 2 points starting at 8 am. After crossing each, A & B took 80 mins & 45 mins to reach their destination. At what time they met with each other?

$\boxed{\text{Ans: 9 am}}$

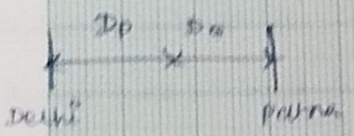
$$t = \sqrt{t_1 t_2} = \sqrt{80 \times 45} = \sqrt{3600} = 60 \text{ min}$$

They met 1hr after 8 am ie 9 am.

26. Two trains P & Q start their journey from delhi & patna respectively towards each other. after crossing each other. Train P & Q takes 25 hrs & 16 hrs to reach their destination. If the speed of train Q is 20 kmph. find the distance

between delhi and patna.

$$\boxed{\text{Ans}: 720\ \text{km}}$$



$$\frac{S_p}{S_a} = \sqrt{\frac{t_a}{t_p}} = \frac{S_p}{20} = \sqrt{\frac{16}{25}}$$

$$\boxed{S_p = 16\ \text{kmph}}$$

$$D = S_p t_p + S_a t_p$$

$$= 16 \times 25 + 20 \times 16$$

$$= 720\ \text{km}$$